Malhar Udmale

Bachelor of Technology
Indian Institute of Information Technology Allahabad

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Education

•Indian Institute of Information Technology, Allahabad *2024-Present*

B.Tech in Information Technology - Business Informatics CGPA: 7.8

Research Publications

•ICML 2026 (International Conference on Machine Learning) *Accepted May 2026*

Co-Author

- Co-authored **"Refusal Behavior Beyond English: A Multilingual Study of Representation Directions"** (Epistemic Intelligence in Machine Learning Workshop, ICML 2026).
- Prior work claims LLM refusal is controlled by **one universal direction**, but tests it only in English. Tested this on English and 5 Indian languages (Qwen 7B/14B/32B).
- Built **IndicRefuse** (6 languages). Ablated a refusal direction per language; scored with a **GPT-4o-mini judge**.
- Ablation removes refusal tokens but gives little harmful compliance in Indic languages (Qwen2-7B Gujarati: refusal 0.34→0.02, compliance 0.12); larger models resist more.

•CVIP 2026 (International Conference on Computer Vision and Image Processing) *Accepted 2026*

Co-First Author

- Co-first-authored **"Federated Multi-Task Learning for Bladder Tumor Segmentation and MIBC Classification Using a Hybrid CNN-Transformer Architecture"**, accepted at CVIP 2026.
- Built a **FedAvg** framework (Flower) across 4 clinical centers on FedBCa (275 T2-weighted MRI scans). Designed **Swin Hybrid**: ResNet-34 + Swin-Tiny encoder with segmentation-guided attention pooling.
- Showed **Geo+Elastic** augmentation gives the best federated result: **DSC 0.8100, patient-level AUC 0.8931**, with no patient data sharing.
- Joint segmentation and MIBC staging in one model: tumor localization guides the staging decision.

Technical Skills and Research Interests

Research Interests: Mechanistic Interpretability, AI Alignment, AI Security, Federated Learning, Medical Imaging, Low-Latency Systems, High-Performance Computing.

Languages: C++, C, Python, Java, SQL.

Systems & CS: C++20, Data Structures & Algorithms, Concurrency, Memory Management, Memory Pools, SPSC Queues, Cache-Aware Design, Intrusive Data Structures, Operating Systems.

Deep Learning: LLMs (Qwen, Gemma), Transformers, CNNs, Sparse Autoencoders, LoRA, PyTorch.

Software & Tools: CMake, Git, GitHub, FastAPI, React, REST APIs, NumPy, Pandas, Scikit-Learn.

Projects

•Low-Latency Stock Exchange Matching Engine

C++20, CMake, Memory Pools, SPSC Queues, Intrusive Data Structures, Cache-Aware Design

- Built a deterministic, header-only **C++20 matching engine** with **price-time priority** across multiple symbols. Supports Limit, Market, IOC, FOK, Stop, Iceberg, and Post-only orders.
- Cut dynamic allocation on the matching path with **intrusive FIFO price levels**, fixed-capacity **memory pools**, and pre-allocated event storage.
- Wrote **lock-free SPSC ring buffers** (acquire/release ordering, cache-line-separated cursors). Used trivially-copyable events, fixed-point prices, and **-fno-exceptions**.

•Smart SAC: Equipment and Room Management Platform

React, TypeScript, Node.js, Express, PostgreSQL, Prisma, Zod, SSE, Docker

- Built a full-stack platform for the IIIT Allahabad Student Activity Center: live equipment status, **15-minute reservations**, room booking, tickets, chat, and a **keyboard-first lend/return desk** with barcode and webcam scan.
- Designed a **PostgreSQL** schema with **Prisma** migrations. Enforced borrowing rules (late returns over 30 days, 7-day blocks, overdue checks) in a service layer; **node-cron** jobs send reminders and expire holds.

Achievements

Adobe University Hackathon 2026 : Top-8 out of 1 lakh+ teams. Built a tool to find website severities and flag them for AI Discoverability.

Amazon ML Challenge 2026 : Scored **95.8%** in Amazon ML Challenge 2026, showing excellent ML domain knowledge

Sarvam AI Campus Hackathon : Runner's Up. Build a voice agent for blind people with multilingual support